

Sergey Krivosheev

Data Engineer

📞 +7-924-153-10-73 | @ krivosheev-sergey-work@yandex.ru | 🔥 FlameInBrainSerj | 🧠 FlameInBrain

EXPERIENCE

Data Engineer

Jul 2024 – Present

Nexign - Leading Russian OSS/BSS software vendor for Billing and Telecom

- Maintained and developed **ETL** pipelines in **Apache Airflow**, including tasks involving interactions with multiple services such as external **APIs**, **HDFS**, **Spark**, **SFTP**, **S3**, etc. Developed plugins for external API integration. Improved validation tasks performance by 3 times
- Migrated **Spark** jobs written in **Java** using **Spark RDD** to **Scala**-based jobs using the **Spark DataFrame API**. Refactored legacy jobs, reducing resource consumption by 30% and execution time by 80% on average. Engineered new functionality using **Spark UDFs**
- Researched and modified **Spark** source code to extend its functionality, including integrating **Spark Streaming** with **SFTP** and modifying the **Spark JDBC** connector to insert nullable strings into enum columns in **PostgreSQL**
- Optimized **ClickHouse** queries in **ETL** jobs, reducing execution time by 30%, and refactored the storage schema of several tables, decreasing read time by an average of 40%
- Researched and integrated **Flink** for implementing **stateful streaming jobs**. Developed jobs in Java using the **DataStream API** with custom sink operators supporting **exactly-once** semantics. Configured deployment for a self-managed job cluster using **Ansible**
- Deployed and administered an **Arenadata Dist Hadoop** cluster in a staging environment with **Hive** and **Spark**. Configured **HDFS** and **Spark** clients on client machines. Migrated **HDFS** data and the **Hive Metastore** from the old cluster to the new one, resolving all incompatibilities
- Successfully completed major versions migrations: **Python** 3.6 → 3.12, **Airflow** 2 → 3, **Spark** 2 → 3, **Java** 8 → 21, **Spring Boot** 2 → 3, **Hadoop** 2 → 3
- Developed several **Kotlin** services as MVPs. Increased **RPS** can be processed by implementing **multi-instance, thread-safe concurrency** with **at-least-once** semantics, enabling horizontal scaling by X times and vertical scaling by 25%. Enhanced transparency by improving service observability, including detailed logs, tracing, and comprehensive monitoring
- Maintained **Java/Kotlin** backend services by fixing bugs in existing functionality and developing new features. Refactored legacy projects with **improving developer experience, reducing time-to-market** for new features by 25%

EDUCATION

Higher School of Economics (HSE University), Faculty of Computer Science Sep 2023 – Jun 2025

Master's degree in Machine Learning and High Load Systems

GPA: 9.87/10

Thesis: Solar Magnetic Tornado Detection Using Neural Networks

Higher School of Economics (HSE University)

Sep 2019 – Jun 2023

Bachelor's degree in Economics

GPA: 8.78/10

SKILLS

Programming Languages: Python, Kotlin, Java, Scala

Data & Data Engineering: SQL, PostgreSQL, ClickHouse, Airflow, Pandas, NumPy, Hadoop Ecosystem (HDFS, YARN, Hive), Spark (Batch & Streaming), Flink, Kafka, S3/Ceph

JVM & Java Frameworks: Java 8/21, Spring Boot (MVC, Security, Data JPA/Hibernate, AOP), Concurrency, JUnit, Mockito, Gradle

Python Frameworks & Tools: FastAPI, Celery, SQLAlchemy, Psycopg, Pytest, Poetry, Flake8, Black, Mypy

Backend & DevOps Tools: Nginx, Redis, Prometheus, Grafana, Telegraf, Docker, RPM, Bash, Linux, WireMock, Ansible, GitLab CI

Languages: English (C1), Russian (Native)